

The Loyola Index Family and the Resolution Limit of Degree Imbalance

Advik Natarajan^a, Ganesh Shiwakoti^b, Michael
Arockiaraj^{a,*}

^a*Department of Mathematics, Loyola College, Chennai, India*

^b*Department of Mathematics and Statistics, Florida Atlantic University,
USA*

marockiaraj@gmail.com

(Received October 9, 2026)

Abstract

The two-parameter Gutman–Milovanović family $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ unifies the Zagreb, Randić, sum-connectivity and related degree-based indices. We study its normalised degree-imbalance exponential extension, the *Loyola index family*

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right),$$

with $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$ identically. Pre-registered tests on measured boiling points show no predictive gain from γ , and a gain on vaporisation enthalpies seen in a narrow search box disappears once the parent is searched over a box that does not truncate it. On the ratio plane, where each edge enters only through its degree ratio, the pure- γ points at algebraic $\gamma \neq 0$ separate exactly the imbalance histograms, a limit shared by every degree-ratio index and lost on trees from order thirteen. We also bound the order exchanges along γ and determine the extremal trees of the pure- γ points.

*Corresponding author.

1 Introduction

Degree-based topological indices form one of the most developed descriptor classes of chemical graph theory. The class runs from the Randić and Zagreb indices [20, 38] and their parametric extensions, the general Randić and sum-connectivity families [6, 47], to the Sombor index [15] and its many variants, surveyed in [16]. Extremal results on bond-incident-degree (BID) indices are surveyed in [2], and a finite-dimensional linear framework for such indices is developed in [35].

The *Gutman–Milovanović family* [18], studied in the recent bounds and QSPR literature [14, 27, 33], is $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$, $\alpha, \beta \in \mathbb{R}$. Its special cases, catalogued in [27], include $M_{0,1} = M_1$, $M_{1,0} = M_2$, the hyper-Zagreb index $M_{0,2} = HM$ [41], $M_{-1,0} = {}^m M_2$, the Randić index $M_{-1/2,0} = R$, the sum-connectivity index $M_{0,-1/2} = \chi$ [46], the harmonic index $2M_{0,-1} = H$, the inverse sum indeg index $M_{1,-1} = ISI$, the geometric–arithmetic index $2M_{1/2,-1} = GA$ [43] and the arithmetic–geometric index $\frac{1}{2}M_{-1/2,1} = AG$. Its two parameter axes are the general Randić and sum-connectivity families, and parameter-optimised versions have already been fitted to octane properties [3].

We study the three-parameter extension

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{\gamma q_{uv}}, \quad q_{uv} = \frac{|d_u - d_v|}{d_u + d_v},$$

the *Loyola index family*, with $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$. Positive γ up-weights degree-imbalanced edges, negative γ downweights them, and on regular graphs γ has no effect. The question is what the imbalance coordinate γ adds to the parent family.

The family is a three-parameter subfamily of the exponential vertex-degree-based indices of Rada [34], whose extremal theory is developed in [8, 13, 21, 42]. Published one-parameter exponential indices [10, 11] place their parameter on a single classical generator. The variable Euler–Sombor index [19] and the cosine-rule Sombor indices [31] weight the imbalance polynomially and without normalisation. The ratio q_{uv} itself appears in the diminished Sombor index [28], in degree-ratio indices [17, 32], and,

as a normalised form of Albertson’s irregularity [1, 29], in $IRLA(G) = 2 \sum_{uv} q_{uv}$ [39]. The pure- γ point $LO(G; 0, 0, \gamma) = \sum_{uv} e^{\gamma q_{uv}}$ is the exponential index, in the sense of [34], of the edge function γq_{uv} . At $\gamma = 2$ it is the sum $\sum_{uv} e^{2q_{uv}}$, which differs from $\exp(IRLA(G))$. What is new is a coordinate linear in q_{uv} in the exponent, coupled with its own parameter to the Gutman–Milovanović family. It makes every edge weight log-linear in (α, β, γ) , so LO is the partition function of an exponential family on the edges of G .

On graphs with $\Delta \leq 4$ every BID index is a linear functional of the ten edge-degree-pair counts [35], so the whole family lies in a fixed ten-dimensional space. Our questions concern a nonlinear family inside that space: which count vectors a parameter point separates, and how the index moves along γ . The tools are classical. Gutman’s survey [16] observes that new variants hardly ever improve on earlier ones in applications, and our tests agree. The family is not proposed as a better predictor, and the QSPR sets used here are small.

Our contributions are as follows.

- (i) The imbalance direction is affinely independent of the parent coordinates once a vertex of degree at least 3 is admitted (Lemma 3.1), and $\log LO$ is convex in (α, β, γ) (Theorem 3.3).
- (ii) The ratio-plane theorem (Theorem 3.5) characterises exactly what the pure- γ points discriminate, with tree collisions at every order from 13, and from 16 for $\Delta \leq 4$. Its corollary extends the collisions to every degree-ratio index, such as GA , AG and the diminished Sombor index.
- (iii) A bound on the number of order exchanges along γ (Proposition 3.11) and per-graph identifiability (Proposition 3.13).
- (iv) The degree-pair ceiling on the octane benchmark (Proposition 4.1), the first tree orders at which discrimination is lost, and structure sensitivity (Section 4).
- (v) Closed forms on standard graphs, the extremal trees of the pure- γ points and termwise bounds (Section 5).

- (vi) A matched-budget QSPR diagnostic and pre-registered tests on measured boiling points (Section 6), in which freeing γ gives no demonstrated gain once the parent’s search box does not truncate it (Section 6.2), and ridge regression on the degree-pair counts leads every single index (Section 6.4).

2 Preliminaries and notation

Let $G = (V, E)$ be a finite simple connected graph of order $n = |V|$ and size $m = |E|$, and for a vertex $v \in V$ let the degree $d_v = d_G(v)$ count the edges incident to v . The maximum and minimum degrees are denoted $\Delta = \Delta(G)$ and $\delta = \delta(G)$. The chemometric sections use hydrogen-suppressed molecular graphs of alkane carbon skeletons, for which $\Delta \leq 4$, while the mathematical results do not assume this bound unless stated.

For an edge $e = uv \in E(G)$ we write (d_u, d_v) for its *degree pair*, understood as an unordered multiset $\{d_u, d_v\}$. For integers $1 \leq i \leq j$ let $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$ be the number of edges with degree pair (i, j) , called (i, j) -edges; the vector $(m_{ij}(G))_{i \leq j}$ is the *degree-pair profile* (or count vector) of G .

Definition 2.1 (Loyola index family). Let $G = (V, E)$ be a simple connected graph. For $\alpha, \beta, \gamma \in \mathbb{R}$, the *Loyola index family* is defined as

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \quad (2.1)$$

Eq. (2.1) is defined for all $\alpha, \beta, \gamma \in \mathbb{R}$ on graphs without isolated vertices. Write $q_{uv} = \text{imb}(d_u, d_v)$, where $\text{imb}(x, y) = |x - y|/(x + y)$, for the normalised degree imbalance of the edge uv , and $q_{ij} = (j - i)/(i + j)$ for $i \leq j$. Each summand depends only on the unordered degree pair of its edge, so

$$LO(G; \alpha, \beta, \gamma) = \sum_{i \leq j} m_{ij}(G) (ij)^\alpha (i + j)^\beta e^{\gamma q_{ij}}. \quad (2.2)$$

In particular $LO(G; \alpha, \beta, 0) = M_{\alpha, \beta}(G)$, the Gutman–Milovanović family [18, 27], and $LO(G; \alpha, \beta, \gamma) = M_{\alpha, \beta}(G)$ for every γ when G is regular.

Definition 2.2 (q -histogram). The q -histogram of G is the function $n_q(G) = \sum_{i \leq j, q_{ij}=q} m_{ij}(G)$, $q \in [0, 1]$, the number of edges of G with imbalance q .

Equal profiles give equal q -histograms. The converse fails (Theorem 3.5(iii)).

3 Parameter geometry

Throughout this section $\theta = (\alpha, \beta, \gamma)$ and $z_{uv} = (\ln(d_u d_v), \ln(d_u + d_v), q_{uv})$, so that

$$LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}. \quad (3.1)$$

For a degree pair (i, j) we write z_{ij} , and call $(1, z_{ij}) \in \mathbb{R}^4$ its augmented feature vector. A set of augmented vectors spans $\mathbb{R}^4$ if and only if the corresponding z_{ij} do not lie in one affine plane of $\mathbb{R}^3$.

Lemma 3.1 (Affine independence of the imbalance direction). *For $\Delta \geq 3$, with $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ the admissible degree pairs, the augmented feature vectors $(1, \ln(ij), \ln(i+j), q_{ij})$, $(i, j) \in P_\Delta$, span $\mathbb{R}^4$. Consequently, if two parameter triples θ, θ' satisfy $e^{\theta^\top z} = c e^{\theta'^\top z}$ for some constant $c > 0$ and all $(i, j) \in P_\Delta$, then $\theta = \theta'$ and $c = 1$. In particular, no choice of (α', β') and scaling constant reproduces the action of $\gamma \neq 0$ on all admissible degree pairs. For $\Delta = 2$ the span has dimension 3, while for $\Delta = 1$ the only pair is $(1, 1)$, with augmented vector $(1, 0, \ln 2, 0)$, and the span has dimension 1. In both cases the conclusion fails. For $\Delta = 2$, indeed, $q_{ij} = [\ln(i+j) - \ln 2 - \frac{1}{2} \ln(ij)] / (3 \ln 3 - \frac{9}{2} \ln 2)$ on the three pairs, so for $\Delta = 2$ the factor $e^{\gamma q_{ij}}$ equals a constant times $(ij)^a (i+j)^b$ on the three pairs, for suitable constants a, b . The effect of γ is therefore reproduced exactly by a shift of (α, β) together with a scaling constant.*

Proof. The vectors $(1, \ln(ij), \ln(i+j))$ on the pairs $(1, 1), (2, 2), (1, 2)$ have rank 3, since the corresponding 3×3 determinant equals $\ln 2 \cdot \ln(9/8) \neq 0$, so it suffices to show that q_{ij} is not an affine function of $(\ln(ij), \ln(i+j))$ on P_Δ . The four pairs $(1, 1), (2, 2), (1, 2), (1, 3)$, all admissible once $\Delta \geq 3$, witness this. The pair $(1, 1)$ occurs only in K_2 , and the conclusion

persists without it, since the augmented vectors of $(1, 2), (1, 3), (2, 2), (3, 3)$, namely $(1, \ln 2, \ln 3, \frac{1}{3}), (1, \ln 3, \ln 4, \frac{1}{2}), (1, \ln 4, \ln 4, 0)$ and $(1, \ln 9, \ln 6, 0)$, have determinant $-\frac{1}{6} \ln \frac{3}{2} \ln(2^{13}/3^8) \approx -0.0150 \neq 0$. Suppose $q_{ij} = a + b \ln(ij) + c \ln(i + j)$ on these pairs. The pairs $(1, 1)$ and $(2, 2)$ have $q = 0$, so they give $a + c \ln 2 = 0$ and $a + (2b + 2c) \ln 2 = 0$, hence $a = -c \ln 2$ and $b = -c/2$. Substituting into the $(1, 2)$ and $(1, 3)$ equations gives $\frac{1}{3} = c(\ln 3 - \frac{3}{2} \ln 2)$ and $\frac{1}{2} = c(\ln 2 - \frac{1}{2} \ln 3)$. In particular $c \neq 0$, since the left side $\frac{1}{3}$ is nonzero. Three times the first minus twice the second gives $0 = c(4 \ln 3 - \frac{13}{2} \ln 2)$, hence $8 \ln 3 = 13 \ln 2$, i.e. $3^8 = 2^{13}$. This is false ($6561 \neq 8192$). Hence the augmented vectors have rank 4. Indeed, a linear relation among the four coordinates on P_Δ with a nonzero coefficient on q_{ij} would express q_{ij} as an affine function of $\ln(ij)$ and $\ln(i+j)$, which we have just excluded. A relation with zero coefficient on q_{ij} is excluded by the rank-3 determinant. If $e^{\theta^\top} z = c e^{\theta'^\top} z$ on P_Δ , then $(\theta - \theta')^\top z_{ij} = \ln c$ for all admissible pairs, so $(\ln c, \theta' - \theta)$ is orthogonal to every augmented vector. Since the augmented vectors span $\mathbb{R}^4$, the only vector orthogonal to all of them is 0, so $\theta = \theta'$ and $c = 1$. For $\Delta = 2$ only the pairs $(1, 1), (1, 2), (2, 2)$ exist. The 3×3 determinant computed at the start of the proof shows that their three augmented vectors have rank 3, and three vectors cannot have more. For $\Delta = 1$ the single augmented vector has rank 1. ■

Remark 3.2 (Independence versus per-graph identifiability). Lemma 3.1 concerns the edge-weight *function* on the full admissible pair set rather than every individual graph. On a single graph G there may be a direction v along which the index merely rescales, that is, $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$ for all t and some constant c . This happens exactly when the realised feature vectors z_{ij} of the edges of G all lie in one plane of $\mathbb{R}^3$, so that they fail to affinely span $\mathbb{R}^3$ (Proposition 3.13). On an r -regular graph, for instance, $LO(G; \theta) = m e^{\theta^\top z_0}$ with $z_0 = (2 \ln r, \ln 2r, 0)$, so the index values determine the single number $\theta^\top z_0$ and nothing else about θ . If the feature vectors z_{ij} realised by G have affine dimension at most 2, some direction of θ , possibly the γ direction, is unidentifiable from $LO(G; \cdot)$ alone. The lemma guarantees that no such degeneracy is forced once all pairs of the molecular class $\Delta \geq 3$ are available. The three parameters are then determined by the edge-weight function on P_Δ . A single graph,

which realises only a few pairs, may fail to determine them.

By (3.1), $\log LO_G$ is a log-sum-exp function of θ .

Theorem 3.3 (Log-convexity). *Fix a graph G with $m \geq 1$ edges, define $F_G(\theta) = \log LO_G(\theta)$ on $\mathbb{R}^3$, and let $\mathbb{E}_\theta$ and Cov_θ denote mean and covariance under the edge probability measure $w_{uv}(\theta) = e^{\theta^\top z_{uv}} / LO_G(\theta)$. Then F_G is convex, with*

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \quad \nabla^2 F_G(\theta) = \text{Cov}_\theta(z) \succeq 0.$$

In particular $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0, 1)$ and $\partial_\gamma^2 F_G = \text{Var}_\theta(q) \geq 0$, so $\log LO$ is non-decreasing and convex in γ , and it is strictly increasing in γ unless every edge of G joins vertices of equal degree. Moreover $\text{Var}_\theta(q) = 0$ at some (equivalently, every) θ if and only if q_{uv} is constant over the edges of G , which covers regular graphs ($q \equiv 0$) and, for example, (Δ, δ) -semiregular bipartite graphs. A bipartite graph is (Δ, δ) -semiregular when every vertex on one side has degree Δ and every vertex on the other side has degree δ , so every edge has degree pair (δ, Δ) . On these every edge carries the same nonzero imbalance, and $\log LO$ is affine and strictly increasing in γ . F_G is strictly convex if and only if the edge feature vectors $\{z_{uv}\}_{uv \in E}$ do not lie in a proper affine subspace of $\mathbb{R}^3$. On a regular graph $F_G(\theta) = \log m + \theta^\top z_0$ is affine and $\text{Cov}_\theta(z) = 0$.

Proof. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k] \mathbb{E}_\theta[z_l]$, the (k, l) -entry of $\text{Cov}_\theta(z)$, which is positive semidefinite. Convexity follows. The measure $w(\theta)$ has full support on E for every θ . Hence $\text{Cov}_\theta(z)$ is singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace. If they do lie in such a subspace, then along the corresponding v the function $t \mapsto F_G(\theta + tv)$ is affine, so F_G is not strictly convex. If they do not, the Hessian is positive definite at every θ , which implies strict convexity. This proves the strictness criterion. Since $q_{uv} \in [0, 1)$, $\mathbb{E}_\theta[q] \in [0, 1)$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r -regular graph all z_{uv} equal $z_0 = (2 \ln r, \ln 2r, 0)$. ■

Corollary 5.5 is the midpoint case of the convexity of F_G .

3.1 The ratio plane, crossings and identifiability

Since $4ij/(i+j)^2 = 1 - q_{ij}^2$, every admissible pair satisfies $\ln(ij) = 2\ln(i+j) + \ln(1 - q_{ij}^2) - \ln 4$, and therefore

$$LO(G; \alpha, \beta, \gamma) = 4^{-\alpha} \sum_{uv \in E(G)} (d_u + d_v)^{2\alpha+\beta} (1 - q_{uv}^2)^\alpha e^{\gamma q_{uv}}. \quad (3.2)$$

With $\kappa = 2\alpha + \beta$, the parameter κ acts only on the degree sum $d_u + d_v$, and α, γ act only on q_{uv} , a function of the degree ratio d_u/d_v .

Definition 3.4 (Ratio plane). The ratio plane is $\Pi = \{(\alpha, \beta, \gamma) \in \mathbb{R}^3 : \beta = -2\alpha\}$, the plane $\kappa = 0$. It contains the pure- γ points $(0, 0, \gamma)$, $GA = 2LO(\frac{1}{2}, -1, 0)$ and $AG = \frac{1}{2}LO(-\frac{1}{2}, 1, 0)$.

Rada [35] shows that no vertex-degree-based index separates trees with equal profiles, and gives sufficient linear criteria for an index to separate all distinct profiles. The theorem below concerns a coarser loss. On Π distinct profiles with equal q -histograms receive equal values, and part (iii) gives the first tree orders at which such profiles occur.

Theorem 3.5 (Ratio plane). *Let G and H be graphs.*

- (i) *On Π , $LO(G; \alpha, -2\alpha, \gamma) = \sum_q n_q(G) ((1 - q^2)/4)^\alpha e^{\gamma q}$ depends on G only through its q -histogram, and G and H agree at every point of Π if and only if $n_q(G) = n_q(H)$ for all q . Otherwise their values differ at every point of Π outside a set of measure zero.*
- (ii) *For every nonzero algebraic γ , $LO(G; 0, 0, \gamma) = LO(H; 0, 0, \gamma)$ if and only if $n_q(G) = n_q(H)$ for all q . On any finite set of graphs the same holds for all but finitely many real γ .*
- (iii) *Let G and H be trees with distinct edge-degree-pair profiles and equal q -histograms, each having a leaf adjacent to a vertex of the same degree $d \geq 2$. Attaching a pendant path of length $L \geq 1$ at such a leaf in each produces trees with the same two properties and the same maximum degree. The profiles change by the same vector, whether or not the chosen leaf edge was the only $(1, d)$ -edge. Consequently, for every $n \geq 13$ (and every $n \geq 16$ among trees with $\Delta \leq 4$) there*

are non-isomorphic trees of order n with distinct degree-pair profiles that no point of Π separates.

Proof. (i) The formula is immediate from (3.2), since setting $\beta = -2\alpha$ removes the degree-sum factor, and it shows that equal histograms give agreement at every point of Π . For the measure-zero clause, suppose the histograms differ. At $\alpha = 0$ the difference is $\sum_q (n_q(G) - n_q(H)) e^{\gamma q}$, an exponential sum with distinct exponents and not all coefficients zero, so it is not identically zero. This uses the first of two standard facts: a nonzero exponential sum with distinct real exponents is not identically zero, because as $\gamma \rightarrow +\infty$ the term with the largest exponent dominates. The difference on Π is therefore a nonzero real-analytic function of (α, γ) , whose zero set has measure zero. This is the second fact: the zero set of a nonzero real-analytic function on $\mathbb{R}^2$ has Lebesgue measure zero [26]. (ii) Agreement for equal histograms is immediate from the formula in (i). Since degrees are integers, every imbalance q is rational, so for algebraic $\gamma \neq 0$ the exponents γq , $q \in \mathbb{Q}$, are distinct algebraic numbers. The Lindemann–Weierstrass theorem [4] says that for distinct algebraic numbers $x_1, \dots, x_k$ the exponentials $e^{x_1}, \dots, e^{x_k}$ are linearly independent over the field of algebraic numbers. Hence the numbers $e^{\gamma q}$ are linearly independent over the algebraic numbers, and the integer coefficients $n_q(G) - n_q(H)$ must all vanish. For real γ , a nonzero exponential sum with k distinct real exponents has at most $k - 1$ real zeros (Proposition 3.11 below), and a finite set of graphs gives finitely many such sums. (iii) The last clause follows from (i) once the trees exist, since equal histograms give agreement on all of Π . The attachment, a device also used by Rada [35] for edge-type vectors, changes the profile of each tree by the same vector, $-(1, d) + (2, d) + (L - 1)(2, 2) + (1, 2)$, so distinct profiles stay distinct and equal histograms stay equal. The new or changed vertices have degree at most 2, so the maximum degree is unchanged when it is at least 2. Trees realising the two order-13 profiles listed in Section 4.2 are the following. Their edge lists confirm the common q -histogram displayed there: one edge of imbalance 0, two of $\frac{1}{5}$, one of $\frac{1}{3}$, three of $\frac{1}{2}$ and five of $\frac{5}{7}$. T_1 : a path $c a b d e \ell$ in which c carries five pendant leaves, b and d carry one pendant leaf each and ℓ is a leaf, so that $d_c = 6$, $d_a = d_e = 2$, $d_b = d_d = 3$; its edges

are $(1, 6)^5, (2, 6), (2, 3)^2, (3, 3), (1, 3)^2, (1, 2)$. T_2 : a path $cbxyz$ in which c carries five pendant leaves, b one and z two, so that $d_c = 6, d_b = d_z = 3, d_x = d_y = 2$; its edges are $(1, 6)^5, (3, 6), (1, 3)^3, (2, 3)^2, (2, 2)$. Both have a leaf adjacent to a vertex of degree $d = 3$. Consider now the order-16 pair with $\Delta \leq 4$. The profile $\{(1, 2)^2, (1, 4)^8, (2, 4)^2, (4, 4)^3\}$ is realised by a path $ABCD$ of vertices of degree 4. Here A and D carry three pendant leaves each, while B and C carry one pendant leaf and one pendant path of length 2 each. The profile $\{(1, 4)^8, (2, 2)^3, (2, 4)^4\}$ is realised by three vertices A, B, C of degree 4. They are joined as Awb through a vertex w of degree 2 and as $Bv_1v_2v_3v_4C$ through a path of four vertices of degree 2. The vertices A, B and C carry three, two and three pendant leaves. Both trees have the q -histogram $\{0^3, \frac{1}{3}^4, \frac{3}{5}^8\}$, since the pairs $(1, 2)$ and $(2, 4)$ have $q = \frac{1}{3}$, the pair $(1, 4)$ has $q = \frac{3}{5}$ and the pairs $(2, 2)$ and $(4, 4)$ have $q = 0$. Both also have a leaf adjacent to a vertex of degree $d = 4$, which gives every larger order. ■

The first sentence of part (ii) is Rada's Lindemann–Weierstrass argument [34], restated in [35, Example 3], applied to the edge function γq_{uv} : two graphs receive the same value for every algebraic $\gamma \neq 0$ exactly when they have equal q -histograms. It is included for completeness. Part (i) follows directly from the factorisation (3.2). The statement for real γ applies the standard fact that a nonzero exponential sum has finitely many real zeros. The colliding families of part (iii) and the corollary below are new. The minimality of orders 13 and 16 is computer-assisted. It rests on an exhaustive enumeration of all trees of orders 2 to 16 in exact rational arithmetic, reproducible with the code of the repository [30].

Corollary 3.6 (Degree-ratio indices). *Let $\rho_{uv} = \min(d_u, d_v) / \max(d_u, d_v)$ and let $I_f(G) = \sum_{uv \in E(G)} f(\rho_{uv})$ for any function f on $(0, 1]$. Since $\rho_{uv} = (1 - q_{uv}) / (1 + q_{uv})$, I_f depends on G only through its q -histogram. Consequently the trees of Theorem 3.5(iii), which exist at every order $n \geq 13$ (and $n \geq 16$ with $\Delta \leq 4$), are not separated by any such index. This applies to GA , AG , the exponential geometric–arithmetic index [10], the diminished Sombor index [28], the symmetric division deg index [44], the inverse symmetric division deg index and its exponential [11], and the*

degree-ratio indices [17, 32].

Remark 3.7. The corollary follows from the identity for ρ_{uv} . Within the class it describes, the pure- γ points at algebraic $\gamma \neq 0$ attain the most possible: by part (ii) they separate every pair of graphs with distinct q -histograms. GA does not (Remark 3.10).

Off Π the same transcendence argument gives the following.

Proposition 3.8 (Parent distinctions survive). *Let $\alpha, \beta \in \mathbb{Q}$ and let $\gamma \neq 0$ be algebraic. For graphs G, H and each imbalance value q write $c_q(G, H) = \sum_{(i,j): q_{ij}=q} (m_{ij}(G) - m_{ij}(H))(ij)^\alpha(i+j)^\beta$. Then $LO(G; \alpha, \beta, \gamma) = LO(H; \alpha, \beta, \gamma)$ if and only if $c_q(G, H) = 0$ for every q . In particular, if $M_{\alpha, \beta}(G) \neq M_{\alpha, \beta}(H)$ then $LO(G; \alpha, \beta, \gamma) \neq LO(H; \alpha, \beta, \gamma)$: at rational exponents a nonzero algebraic γ never destroys a distinction made by the parent.*

Proof. The difference is $\sum_q c_q(G, H) e^{\gamma q}$, a sum over the distinct imbalances q occurring in G or H . For rational α, β the numbers $(ij)^\alpha(i+j)^\beta$ are algebraic, being rational powers of positive integers, so the coefficients $c_q(G, H)$, integer combinations of them, are algebraic. For algebraic $\gamma \neq 0$ the exponents γq are distinct algebraic numbers, so by the Lindemann–Weierstrass theorem [4] the sum vanishes only if every $c_q(G, H)$ vanishes. The last claim follows because $\sum_q c_q(G, H) = M_{\alpha, \beta}(G) - M_{\alpha, \beta}(H)$. ■

Remark 3.9. Two graphs merged by $M_{\alpha, \beta}$ stay merged at an algebraic $\gamma \neq 0$ only if every class sum $c_q(G, H)$ vanishes, which is stronger than $M_{\alpha, \beta}(G) = M_{\alpha, \beta}(H)$. On the trees of order 16, $\gamma = 1$ recovers the distinctions the parent loses at $(0, \frac{1}{2}, 0)$ and $(0, \frac{1}{3}, 0)$ (Section 4.2). The algebraic hypothesis cannot be dropped: at $\gamma = \ln 4$, $LO(K_4; 0, 0, \gamma) = 6 = 3e^{\gamma/2} = LO(K_{1,3}; 0, 0, \gamma)$.

Remark 3.10 (The floor on Π). By exhaustive enumeration of all trees of orders 2–17, two trees of order at most 12 with equal q -histograms have equal degree-pair profiles, and the same holds for trees with $\Delta \leq 4$ of order at most 15. With part (iii), the pure- γ points at algebraic $\gamma \neq 0$ attain the degree-pair floor of Section 4.1, separation of all distinct profiles, exactly up to those orders and at no larger

order. At orders 13–17 generic points off Π (Proposition 4.2) and the Randić index separate more trees than any point of Π . A point of Π can also do worse. GA merges two distinct q -histograms among trees first at order 17, for instance $\{(1, 5)^{10}, (2, 2)^2, (5, 5)^2, (2, 5), (1, 2)\}$ and $\{(4, 5)^3, (1, 4)^5, (1, 5)^6, (2, 5), (1, 2)\}$, which have equal GA and different q -histograms.

Proposition 3.11 (Order exchanges along γ). *Fix (α, β) and graphs G, H , let $q_1 < \dots < q_r$ be the distinct imbalances of the degree pairs occurring in G or H , and write $D(\gamma) = LO(G; \alpha, \beta, \gamma) - LO(H; \alpha, \beta, \gamma) = \sum_{k=1}^r c_k e^{\gamma q_k}$ with $c_k = \sum_{q_{ij}=q_k} (m_{ij}(G) - m_{ij}(H))(ij)^\alpha (i+j)^\beta$. Let V be the number of sign changes in $(c_1, \dots, c_r)$ after zero terms are removed. If $D \not\equiv 0$, then D has at most $V \leq r - 1$ real zeros counted with multiplicity, and their number has the parity of V . In particular, for graphs with $\Delta \leq 4$ the possible imbalances are $0, \frac{1}{7}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}$, and two such graphs exchange order at most five times as γ ranges over $\mathbb{R}$, for every (α, β) .*

Proof. With $x = e^\gamma > 0$, D becomes the generalised polynomial $\sum_k c_k x^{q_k}$ with distinct real exponents, so Laguerre’s extension of Descartes’ rule of signs applies [22]. Consider a finite sum $\sum_k c_k x^{q_k}$ with real exponents $q_1 < \dots < q_r$ and let V count the sign changes in its coefficient sequence. The rule says that the number of zeros in $x > 0$, counted with multiplicity, is at most V and differs from V by an even number. Zeros of D in γ correspond one-to-one, with multiplicity, to positive zeros in x . The number of real zeros of D , with multiplicity, is therefore at most V and has its parity. As an independent check, the parity also follows from the signs of D as $\gamma \rightarrow \mp\infty$, which are the signs of the first and last nonzero c_k . Divide D by $e^{\gamma q_k}$ for the first k with $c_k \neq 0$ and let $\gamma \rightarrow -\infty$, or for the last such k and let $\gamma \rightarrow +\infty$. As in the proof of Proposition 3.17 below, only c_k survives. ■

Remark 3.12 (Sharpness). The bound is attained. The trees of order 12 with profiles $\{(1, 3)^2, (1, 4)^2, (1, 6)^5, (3, 4), (4, 6)\}$ and $\{(1, 2), (1, 5)^4, (1, 6)^4, (2, 6), (5, 6)\}$ have $\Delta = 6$, eight distinct imbalances and $V = 5$, and at $(\alpha, \beta) = (-1, 0)$ they cross at $\gamma \approx -17.2, -12.0, -2.0, 2.7$ and 31.1 , which are therefore all zeros of D . The first tree is a vertex of degree 6 with five

leaves and a neighbour of degree 4, which carries two leaves and a vertex of degree 3 with two leaves. The second is a vertex of degree 6 with four leaves, a neighbour of degree 5 with four leaves, and a neighbour of degree 2 ending in a leaf. Whether five crossings are attained for $\Delta \leq 4$ is open. Along a line $\theta + tv$ not parallel to the γ axis the exponents of D are the values $v^\top z_{ij}$, of which there can be as many as admissible pairs, so the constant five is specific to the γ axis.

For a graph G let $R(G)$ be the set of degree pairs it realises, $r(G)$ the number of distinct imbalances among them, and $A(G)$ the affine dimension of $\{z_{ij} : (i, j) \in R(G)\}$, that is, the dimension of the smallest affine subspace of $\mathbb{R}^3$ containing it.

Proposition 3.13 (Per-graph identifiability). *(i) $A(G)$ equals the rank of the Hessian $\text{Cov}_\theta(z)$ of $\log LO_G$ at every θ , and the directions v for which there is a constant $c = c(v)$ with $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$ for all t and θ form a subspace of dimension $3 - A(G)$.*

(ii) $A(G) \leq \min(|R(G)| - 1, r(G), 3)$ for every graph.

(iii) If $r(G) = 2$ with imbalances $q_1 \neq q_2$, then $v = (1, -2, c)$ with $c = [\ln(1 - q_2^2) - \ln(1 - q_1^2)]/(q_1 - q_2)$ is such a direction.

Proof. (i) The measure $w(\theta)$ puts positive weight on every edge, so, as in the proof of Theorem 3.3, $\text{Cov}_\theta(z)$ is singular exactly in the directions v with $v^\top z$ constant on $R(G)$. Such v are those orthogonal to the differences of the z_{ij} , and they form a subspace of dimension $3 - A(G)$, so the rank of $\text{Cov}_\theta(z)$ is $A(G)$. (In the language of exponential families [7], $w(\theta)$ is the family with sufficient statistic z on the support $R(G)$. The rank statement says that its minimal representation has $A(G)$ parameters.) A direction v rescales the index, $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$ for all θ and t , exactly when $v^\top \nabla F_G(\theta) = c$ is independent of θ . Differentiating, this holds exactly when $\text{Cov}_\theta(z)v = 0$ for all θ , that is, when $v^\top z$ is constant on $R(G)$. The rescaling directions are therefore the subspace of dimension $3 - A(G)$ just described. (ii) By (3.2), $z_{ij} = (2s + \ln(1 - q^2) - \ln 4, s, q)$ with $s = \ln(i + j)$, so pairs of equal imbalance lie on one line parallel to $(2, 1, 0)$, and r parallel lines span an affine space of dimension at most r .

(iii) $v^\top z_{ij} = \ln(1 - q_{ij}^2) - \ln 4 + c q_{ij}$ takes the same value at q_1 and q_2 by the choice of c . ■

Remark 3.14 (The case $\Delta = 7$). Equality holds in (ii) for every set of admissible pairs with $\Delta \leq 6$ (by exhaustive enumeration), hence for all molecular graphs. It fails from $\Delta = 7$: among the subsets of P_7 with at most four pairs, the only one violating equality is $\{(1, 7), (2, 6), (3, 5), (4, 4)\}$, whose degree sums all equal 8. Every violating set contains a violating subset of at most four pairs, namely three collinear points with two imbalances or four coplanar points with three imbalances. Any fifth pair of P_7 has a degree sum different from 8 and leaves that plane, so this is the only violating set. A constant degree sum makes $\ln(i + j)$ constant on the set, so $A \leq 2$, while the bound equals 3 as soon as the set has at least four pairs and three imbalances. In a connected graph whose edges all have degree sum s , each neighbour of a vertex of degree d has degree $s - d$, so only the pair $\{d, s - d\}$ occurs. Hence no connected graph realises this set.

Example 3.15. The spider with four legs of length three (a centre vertex joined to four paths on three vertices) realises $(2, 4)$, $(2, 2)$ and $(1, 2)$, with the two imbalances $\frac{1}{3}$ and 0. On it γ is confounded with (α, β) along the direction of part (iii).

Remark 3.16 (Data sets). For a QSPR data set the relevant set is the union of the pairs realised by its molecules. A direction that rescales every molecule's index by the same factor leaves every correlation unchanged and is absorbed into a regression slope. On the octanes this union contains nine of the ten admissible pairs and its affine dimension is 3, so no such direction exists, and the fits of Section 6 on the octanes are free of exact confounding of this kind.

Proposition 3.17 (Limits along the γ axis). *Let $E_{\max}(G)$ denote the set of edges realising $\text{imb}_{\max}(G) = \max_{uv} \text{imb}(d_u, d_v)$ and $E_{\min}(G)$ the set realising $\text{imb}_{\min}(G) = \min_{uv} \text{imb}(d_u, d_v)$. Then as $\gamma \rightarrow +\infty$,*

$$\frac{LO(G; \alpha, \beta, \gamma)}{e^{\gamma \text{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

and as $\gamma \rightarrow -\infty$,

$$LO(G; \alpha, \beta, \gamma) / e^{\gamma \text{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta.$$

Proof. With $\mu(G) := \text{imb}_{\max}(G)$ and $\varepsilon_{uv} := \mu(G) - \text{imb}(d_u, d_v) \geq 0$, we have $LO(G; \alpha, \beta, \gamma) / e^{\gamma \mu(G)} = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{-\gamma \varepsilon_{uv}}$. The summands over $E_{\max}(G)$ ($\varepsilon_{uv} = 0$) do not depend on γ , while the others ($\varepsilon_{uv} > 0$) tend to 0 as $\gamma \rightarrow +\infty$, termwise since $E(G)$ is finite. The case $\gamma \rightarrow -\infty$ is dual. $\blacksquare$

4 Discrimination

4.1 The degree-pair ceiling

No vertex-degree-based index separates graphs with equal degree-pair counts, so the number of distinct profiles is a *ceiling* on distinct values and a *floor* on degeneracy.

Proposition 4.1 (Degree-pair ceiling). *If two graphs G and H have identical unordered edge-degree-pair count vectors, $m_{ij}(G) = m_{ij}(H)$ for all $i \leq j$, then $LO(G; \theta) = LO(H; \theta)$ for every $\theta \in \mathbb{R}^3$, and more generally $TI(G) = TI(H)$ for every vertex-degree-based index TI . Among the 18 octane isomers exactly 16 distinct count vectors occur, and the pairs $\{3\text{-methylheptane}, 4\text{-methylheptane}\}$ and $\{3,4\text{-dimethylhexane}, 3\text{-ethyl-2-methylpentane}\}$ share their count vectors but differ in measured properties (Table 1). Hence every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible error on this benchmark.*

Proof. By Definition 2.1, $LO(G; \theta)$ depends on G only through the multiset of unordered edge degree pairs, that is, through $(m_{ij}(G))_{i \leq j}$, and the same holds for every vertex-degree-based index [35]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package, and Table 1 lists the common count vectors of both collision pairs. $\blacksquare$

The ceiling is tight. Distinct count vectors are separated at almost every parameter choice.

Proposition 4.2 (Generic discrimination). *Fix $\Delta \geq 1$ and let $m \neq m'$ be two distinct edge-degree-pair count vectors supported on the admissible pairs P_Δ . Then $D(\theta) = \sum_{(i,j) \in P_\Delta} (m_{ij} - m'_{ij}) e^{\theta^\top z_{ij}}$ is a real-analytic function of θ that is not identically zero, and its zero set has Lebesgue measure zero in $\mathbb{R}^3$ [26]. Consequently, any two graphs with distinct count vectors are distinguished by $LO(\cdot; \theta)$ for almost every $\theta \in \mathbb{R}^3$, while only graphs with identical count vectors (Proposition 4.1) are indistinguishable for all θ .*

Proof. The map $(i, j) \mapsto z_{ij}$ is injective on P_Δ , since $\ln(ij)$ and $\ln(i + j)$ recover the product and the sum, which determine the unordered pair. Choose $v \in \mathbb{R}^3$ such that the inner products $v^\top z_{ij}$ are pairwise distinct, which excludes only finitely many hyperplanes of v . Along the ray $\theta = tv$, $D(tv) = \sum_k c_k e^{t\lambda_k}$ with distinct exponents λ_k and coefficients $c_k = m_{i_k j_k} - m'_{i_k j_k}$ that are not all zero, and such an exponential sum vanishes identically only if all $c_k = 0$. Hence $D \neq 0$ on the ray, and so on $\mathbb{R}^3$. Since D is a finite sum of exponentials of linear functionals, it is real-analytic on $\mathbb{R}^3$, and the zero set of a nonzero real-analytic function has Lebesgue measure zero. ■

In the terminology of Rada, Rodríguez and Sigarreta [36], an index is weakly discriminating over a class when equal values force equal count vectors; they showed that several classical indices fail this over chemical trees while the Sombor index satisfies it. Proposition 4.2 says that almost every point of the family is weakly discriminating over every class.

The proof uses only the injectivity of $(i, j) \mapsto (\ln(ij), \ln(i + j))$ and therefore applies to the parent $M_{\alpha, \beta}$ on the (α, β) -plane, so Proposition 4.2 is inherited from the parent rather than conferred by γ . Since $q_{ij} = \sqrt{1 - 4ij/(i + j)^2}$ depends only on the pair, γ adds no graph information beyond the edge-degree-pair counts, of which every vertex-degree-based index is a functional [35]. What it adds is the functional direction of Lemma 3.1, a different weighting of the same information.

Proposition 4.2 concerns generic real exponents. The indices used in practice have rational exponents, and there the parent can fail: the sum-connectivity index $\chi = M_{0,-1/2}$ and the index $M_{0,1/2}$ with edge weight $\sqrt{d_u + d_v}$ are not weakly discriminating over chemical trees [36, Example 2.2]. The imbalance coordinate separates the pair of trees used there for χ .

Example 4.3. For $t \geq 1$ the chemical trees S and T of order $17 + t$ in Example 2.2 of [36] have $m(S) - m(T) = 6(1, 3) + 4(2, 3) + 4(3, 3) - 4(1, 4) - 6(2, 2) - 4(2, 4)$ and equal sum-connectivity index. At $(\alpha, \beta) = (0, -\frac{1}{2})$,

$$D(\gamma) = LO(S) - LO(T) = 3(e^{\gamma/2} - 1) - \frac{4}{\sqrt{5}}(e^{3\gamma/5} - e^{\gamma/5}) + \frac{4}{\sqrt{6}}(1 - e^{\gamma/3}),$$

so $D(0) = 0$ and $D(1) \approx 0.2255$. The separation is small for small γ , for instance $D(10^{-6}) \approx 2.4 \times 10^{-7}$.

Table 1. The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector, shown as multiplicities of the unordered pairs, and is therefore indistinguishable by every vertex-degree-based index for all parameter values. The two members differ in the measured properties. Units: T_B in $^{\circ}\text{C}$; ΔH_f and ΔH_{vap} in kcal mol^{-1} , S in $\text{cal mol}^{-1} \text{K}^{-1}$; ω is dimensionless.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
<i>shared count vector</i> 2(1, 2), 1(1, 3), 2(2, 2), 2(2, 3):					
3-methylheptane	118.9	-50.82	9.52	111.26	0.371
4-methylheptane	117.7	-50.69	9.48	109.32	0.372
<i>shared count vector</i> 2(1, 2), 2(1, 3), 2(2, 3), 1(3, 3):					
3,4-dimethylhexane	117.7	-50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	-50.48	9.21	106.06	0.330

4.2 Degeneracy across tree orders

Exponential indices of this type were introduced by Rada [34] for their discriminating power on trees.

Degeneracy [23] measures the failure to discriminate non-isomorphic graphs, and we quantify it with the distinct-value convention $100(1 -$

k/N), where k is the number of distinct values among $N = 106$ trees (Supplementary Section S5). On the 106 trees of order 10 the pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ attain 25.5% (79 distinct values), tied with R , GA and AG . The 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the floor for *any* degree-pair-based index, while M_1 (83.0%) and M_2 (67.0%) are the most degenerate (Supplementary Figure S2). For orders 7 through 12, on all trees and on those with $\Delta \leq 4$ (Supplementary Table S4), the pure- γ points and R attain the floor at every order. The index χ leaves it from order 10, with 77 values against 79 profiles. Theorem 3.5 makes this exact. The pure- γ points separate precisely the q -histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for $\Delta \leq 4$) and not beyond. At order 13 the profiles $\{(1,2)^1, (1,3)^2, (1,6)^5, (2,3)^2, (2,6)^1, (3,3)^1\}$ and $\{(1,3)^3, (1,6)^5, (2,2)^1, (2,3)^2, (3,6)^1\}$ (exponents are edge multiplicities) share the q -histogram $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$. Table 2 collects the first failures, and the index R still attains the floor at order 13 (570 values) and leaves it at order 14, where the pure- γ points give 1090 values.

Table 2. First tree order at which discrimination is lost. Values are counted as distinct at 11 significant digits. “Below floor” is the first order with fewer distinct values than distinct edge-degree-pair profiles, the floor for every vertex-degree-based index. The entry gives (values/profiles) at that order. “Merges” is the first order at which an index on Π gives equal values to two trees with different q -histograms. This does not apply to R , which is not on Π . $LO(0,0,\gamma)$ denotes the pure- γ points $\gamma = 1, 2$. Orders below 7 were not examined for the value counts. Equal q -histograms of distinct profiles were sought at every order from 2 and first found at the orders shown.

Index	all trees		trees with $\Delta \leq 4$	
	below floor	merges	below floor	merges
$LO(0,0,\gamma)$	13 (566/570)	never ^a	16 (809/810)	never ^a
R	14 (1095/1100)	n/a ^b	17 (1191/1194)	n/a ^b
GA	13 (566/570) ^d	17 ^c	16 (809/810) ^d	> 17 ^e

^a All q -histograms are separated for every nonzero algebraic γ (Theorem 3.5(ii)).

^b R is not on Π . ^d GA lies on Π and merges no q -histograms below order 17, so there it ties the pure- γ points. ^e The witness (Section 3.1) has maximum degree 5; among trees with $\Delta \leq 4$, GA merges no q -histograms up to order 17, the largest examined.

Off the ratio plane the imbalance factor can restore discrimination lost at particular classical exponents. A point $(0, \beta, 0)$ weights an edge by its degree sum only, so pairs with equal sum receive equal weight. With $\gamma \neq 0$ their imbalances separate them. On all trees of order 16 (4069 profiles), $LO(0, \frac{1}{2}, 0)$ and $LO(0, \frac{1}{3}, 0)$ take only 3475 and 3476 distinct values, while $LO(0, \frac{1}{2}, 1)$ and $LO(0, \frac{1}{3}, 1)$ separate all 4069. $LO(\frac{1}{2}, 0, \cdot)$ rises from 3940 to 4067 and the Randić point $LO(-\frac{1}{2}, 0, \cdot)$ from 4007 to 4068 as γ goes from 0 to 1. Points on Π such as GA are unaffected (3996 in both cases), consistent with Theorem 3.5, and generic parent points such as $LO(0.3, 0.2, 0)$ already separate all 4069 profiles (Proposition 4.2). The effect is established by enumeration for orders 12–16 but is not proved. It yields no predictive gain over a tuned parent in Section 6, and at fixed fractional exponents the sign of the change depends on the property (Supplementary Section S5).

4.3 Structure sensitivity

The structure-sensitivity criterion of Furtula, Gutman and Dehmer [12], with a fingerprint-based variant due to Rakić and Furtula [37], was implemented with the published protocol on the 75 decane isomer trees. For a tree, the similar structures are those obtained by relocating a single edge within the molecular class. SS and Abr are the class means of the mean and the maximum relative change of the index over these neighbours. The implementation reproduces the published decane-class values of Barman and Das [5], with eleven of fourteen SS/Abr values agreeing exactly at the four displayed decimals. Each of the other three agrees within one unit in the fourth. The protocol and this control are detailed in Supplementary Section S6. Under the published desideratum (SS large, Abr small), $LO(0, 0, 2)$ has a structure sensitivity indistinguishable at three decimals from that of the hyper-Zagreb index (0.1257 against 0.1256). Its abruptness is markedly smaller (0.274 against 0.313). The sensitivity-to-abruptness ratios of the pure- γ points (0.4579 and 0.4565) are comparable to those of $H/2$, GA and R (0.448–0.451, Table 3). In view of the redundancy noted in Section 6.1, these third-decimal differences do not rank the indices. SS is also not a fixed property of the family. Along the pure- γ line it grows with γ (0.031, 0.062, 0.126, 0.319 at $\gamma = 0.5, 1, 2, 5$), while the

ratio peaks near $\gamma = 2$ and then falls (0.427 at $\gamma = 5$). The pure- γ points reduce to first order in γ to the parameter-free aggregate *IRLA*, since $LO(0, 0, \gamma) = m + \frac{\gamma}{2} IRLA + O(\gamma^2)$, and this aggregate has a higher ratio ($SS = 0.2343$, $Abr = 0.5066$, ratio 0.4625). The higher ratio of this first-order aggregate shows that the favourable sensitivity–abruptness balance is a property of linear imbalance weighting rather than of the exponential form. The equality of SS with hyper-Zagreb at $\gamma = 2$ reflects the chosen value of γ .

Table 3. Structure sensitivity SS , abruptness Abr , and their ratio on the 75 decane isomers under the published similar-structure protocol of [12] (single-edge relocation within the molecular class).

Index	SS	Abr	SS/Abr
M_1	0.0566	0.1336	0.4238
M_2	0.0922	0.2443	0.3776
HM	0.1256	0.3126	0.4017
mM_2	0.0516	0.1177	0.4384
R	0.0264	0.0588	0.4481
χ	0.0264	0.0591	0.4463
$H/2$	0.0516	0.1143	0.4514
ISI	0.0226	0.0597	0.3795
GA	0.0241	0.0535	0.4507
AG	0.0256	0.0579	0.4415
$LO(0, 0, 1)$	0.0624	0.1367	0.4565
$LO(0, 0, 2)$	0.1257	0.2745	0.4579

5 Closed forms, bounds and extremal trees

5.1 Closed-form values on standard graphs

The classical graph families used are the complete graph K_n , the cycle C_n , the path P_n , the star S_n and the complete bipartite graph $K_{p,q}$. Beyond these, the hypercube Q_k has 2^k vertices and $k \cdot 2^{k-1}$ edges and is k -regular. The wheel W_n is an $(n-1)$ -cycle plus a universal vertex, and the friendship graph F_p consists of p triangles sharing a common vertex. We also treat r -regular graphs.

Proposition 5.1 (Closed forms). *Let $\alpha, \beta, \gamma \in \mathbb{R}$. Then:*

1. $LO(K_n; \alpha, \beta, \gamma) = \frac{n(n-1)}{2} 2^\beta (n-1)^{2\alpha+\beta}$ for $n \geq 2$.

2. $LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta}$ for $n \geq 3$.

3. $LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^\beta, & n = 2, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$

4. $LO(S_n; \alpha, \beta, \gamma) = (n-1)(n-1)^\alpha n^\beta \exp\left(\gamma \frac{n-2}{n}\right)$ for $n \geq 2$.

5. $LO(K_{p,q}; \alpha, \beta, \gamma) = pq(pq)^\alpha (p+q)^\beta \exp\left(\gamma \frac{|p-q|}{p+q}\right)$.

6. $LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^\beta k^{2\alpha+\beta}$ for $k \geq 1$.

7. For $n \geq 5$, W_n has $n-1$ spoke edges with degree pair $(3, n-1)$ and $n-1$ rim edges with degree pair $(3, 3)$, so

$$LO(W_n; \alpha, \beta, \gamma) = (n-1) \left[3^\alpha (n-1)^\alpha (n+2)^\beta \exp\left(\gamma \frac{n-4}{n+2}\right) + 9^\alpha 6^\beta \right].$$

8. For $p \geq 1$, F_p has $2p$ spoke edges $(2, 2p)$ and p outer edges $(2, 2)$, so

$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p(4p)^\alpha (2p+2)^\beta \exp\left(\gamma \frac{p-1}{p+1}\right).$$

9. For an r -regular graph G with $r \geq 1$ and m edges, $LO(G; \alpha, \beta, \gamma) = m \cdot 2^\beta r^{2\alpha+\beta}$.

Proof. Each case follows by enumerating the edge degree pairs and applying Definition 2.1. In K_n , C_n , Q_k and r -regular graphs every edge has $d_u = d_v = r$, so $\text{imb} = 0$ and $(d_u d_v)^\alpha (d_u + d_v)^\beta = 2^\beta r^{2\alpha+\beta}$, which gives items (1), (2), (6) and (9). All $n-1$ edges of S_n have degree pair $(1, n-1)$ with $\text{imb} = (n-2)/n$ (item (4)), and all pq edges of $K_{p,q}$ have pair (p, q) (item (5)). For $n \geq 4$, P_n has two pendant edges $(1, 2)$ and $n-3$ interior edges $(2, 2)$, and the cases $n = 2, 3$ are immediate (item (3)). Items (7) and (8) follow by summing over the two edge-degree-pair classes of W_n and F_p . ■

5.2 Extremal trees for the pure-imbalance points

Proposition 5.2 (Extremal trees of the pure- γ points). *Let $n \geq 4$ and $\gamma > 0$. Among all trees of order n , the path P_n is the unique minimiser and the star S_n the unique maximiser of $LO(\cdot; 0, 0, \gamma)$, with*

$$(n - 3) + 2e^{\gamma/3} \leq LO(T; 0, 0, \gamma) \leq (n - 1)e^{\gamma(n-2)/n}.$$

For $\gamma < 0$ the roles of P_n and S_n are exchanged.

Proof. Let $\gamma > 0$. Every edge uv of a tree of order n has $q_{uv} \leq (n - 2)/n$, with equality if and only if $\{d_u, d_v\} = \{1, n - 1\}$, so $LO(T; 0, 0, \gamma) \leq (n - 1)e^{\gamma(n-2)/n}$ with equality if and only if every edge is a $(1, n - 1)$ -edge, that is, $T = S_n$. For the lower bound, a tree with $k \geq 2$ leaves has at least k edges incident to a leaf, and each such edge has $q \geq \frac{1}{3}$ because the other end has degree at least 2 and $(d - 1)/(d + 1)$ is increasing in d . Hence $LO(T; 0, 0, \gamma) \geq (n - 1 - k) + ke^{\gamma/3} \geq (n - 3) + 2e^{\gamma/3}$, since $e^{\gamma/3} > 1$. Equality in the second inequality forces $k = 2$, so $T = P_n$, and P_n attains the bound because its two pendant edges are $(1, 2)$ -edges and its other edges are $(2, 2)$ -edges. For $\gamma < 0$ every inequality reverses. ■

Cruz and Rada [9] give general criteria for the path and the star to be extremal among trees; the direct proof above is included because it is short. The star half also follows from the sufficient condition of [35, Section 4.2], since for $\gamma > 0$ the $(1, n - 1)$ -edge has the largest weight $e^{\gamma q}$. The path half does not follow from the corresponding condition there, because the $(1, 2)$ -weight $e^{\gamma/3}$ exceeds the $(2, 3)$ -weight $e^{\gamma/5}$. The proof uses only the bounds on q and does not extend verbatim to general (α, β) , where the ordering of trees depends on the scale exponent $2\alpha + \beta$ as well as on γ . Among chemical trees ($\Delta \leq 4$) the termwise bound gives $LO(T; 0, 0, \gamma) \leq (n - 1)e^{3\gamma/5}$ for $\gamma > 0$, with equality only for $n = 5$. The path remains the unique minimiser for $\gamma > 0$ among chemical trees by Proposition 5.2, while the maximal chemical trees for $n > 5$ are left open. The orders 4 to 12 were also checked by enumeration in the reproducibility package.

5.3 Termwise bounds

Throughout this subsection G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$. For $\gamma \geq 0$ every edge satisfies $1 \leq e^{\gamma q_{uv}} \leq e^{\gamma(\Delta-1)/(\Delta+1)}$. At $\gamma = 0$ bounds of the types below, with their equality cases, are known [14, 27, 33]. The statements below extend them to $\gamma \neq 0$ and track how γ changes the equality cases. Several are machine-checked (Supplementary Section S7).

Definition 5.3 (Edge weight). For real parameters α, β, γ and positive reals x, y ,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\left(\gamma \frac{|x - y|}{x + y}\right). \quad (5.1)$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition 5.4 (Bracket bound). *Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \quad (5.2)$$

where $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality if and only if every edge of G has a degree pair attaining $\psi_{\max}$. The lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Every edge uv has its degree pair in the admissible set $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$, hence $\psi_{\min} \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$ on every edge, and summing over the m edges and using $\sum_{uv} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives (5.2). Each side of (5.2) is the sum of m per-edge inequalities, and is therefore an equality if and only if every per-edge inequality is. That is, equality holds if and only if every edge has a degree pair attaining $\psi_{\max}$ (upper side), respectively $\psi_{\min}$ (lower side), over $\mathcal{P}(\delta, \Delta)$. ■

Corollary 5.5 (Cauchy–Schwarz interpolation). *Let $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, and $\gamma_1 + \gamma_2 = 2\gamma$. Then for every simple connected graph G with*

$m \geq 1$ edges,

$$LO(G; \alpha, \beta, \gamma) \leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) \cdot LO(G; \alpha_2, \beta_2, \gamma_2)}$$

Equality holds if and only if the ratio

$$\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1) / \psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$$

is the same for every edge of G .

Proof. For each edge $e = uv$ write $\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) > 0$, $k \in \{1, 2\}$, and by additivity of the exponents and the hypotheses on $(\alpha_k, \beta_k, \gamma_k)$, $\psi_e^{(1)} \psi_e^{(2)} = \psi(d_u, d_v; \alpha, \beta, \gamma)^2$ on every edge. The Cauchy–Schwarz inequality for $a_e = (\psi_e^{(1)})^{1/2}$ and $b_e = (\psi_e^{(2)})^{1/2}$ gives $\sum_e a_e b_e \leq (\sum_e a_e^2)^{1/2} (\sum_e b_e^2)^{1/2}$, which is the claim because $\sum_e a_e b_e = LO(G; \alpha, \beta, \gamma)$, $\sum_e a_e^2 = LO(G; \alpha_1, \beta_1, \gamma_1)$ and $\sum_e b_e^2 = LO(G; \alpha_2, \beta_2, \gamma_2)$. Equality holds if and only if $b_e = \lambda a_e$ for a constant $\lambda \geq 0$ and all e , i.e. if and only if $\psi_e^{(2)} / \psi_e^{(1)} = \lambda^2$ is the same on every edge. (The inequality itself is midpoint log-convexity of $\theta \mapsto LO(G; \theta)$, Theorem 3.3.) $\blacksquare$

Proposition 5.6 (Upper bound in m and Δ). *Let G be a connected graph of size m and maximum degree Δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \leq m \cdot \Delta^{2\alpha} \cdot (2\Delta)^\beta \cdot \exp\left(\gamma \frac{\Delta - 1}{\Delta + 1}\right). \quad (5.3)$$

For $\Delta = 1$ the graph is K_2 and the bound is attained for every (α, β, γ) . For $\Delta \geq 2$, equality holds if and only if $\alpha = \beta = \gamma = 0$, or $\gamma = 0$, $\alpha + \beta > 0$ and G is Δ -regular, or $\gamma > 0$, $\alpha = \beta = 0$ and $G \cong K_{1, \Delta}$; for $\Delta \geq 2$, $\gamma > 0$ and $\alpha + \beta > 0$ the inequality is strict.

Proof. On every edge uv we have $1 \leq d_u, d_v \leq \Delta$, hence $d_u d_v \leq \Delta^2$ and $d_u + d_v \leq 2\Delta$, and the normalised imbalance $\text{imb}(x, y) = |x - y| / (x + y)$ is maximised over $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$ at $(1, \Delta)$, so $|d_u - d_v| / (d_u + d_v) \leq (\Delta - 1) / (\Delta + 1)$. Since $\alpha, \beta, \gamma \geq 0$, the maps $t \mapsto t^\alpha$, $t \mapsto t^\beta$ and $t \mapsto e^{\gamma t}$ are non-decreasing, so $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \Delta^{2\alpha} (2\Delta)^\beta \exp(\gamma(\Delta - 1) / (\Delta + 1))$,

and summing over the m edges yields (5.3). For $\gamma = 0$ the exponential factor is 1. If $\alpha = \beta = 0$ both sides equal m , and if $\alpha + \beta > 0$ equality forces $d_u d_v = \Delta^2$ or $d_u + d_v = 2\Delta$ on every edge, that is $d_u = d_v = \Delta$, so G is Δ -regular; the converse is immediate. For $\gamma > 0$ and $\Delta \geq 2$ the bound is attained if and only if $\alpha = \beta = 0$ and every edge has degree pair $(1, \Delta)$ (for connected G , $G \cong K_{1,\Delta}$). If $\alpha > 0$ or $\beta > 0$ the three factor-wise maxima are not simultaneously attainable and the inequality is strict. ■

Proposition 5.7 (Lower bound in m and δ). *Let G be a connected graph of size m and minimum degree δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \geq m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \quad (5.4)$$

with equality if and only if G is δ -regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. On every edge $d_u, d_v \geq \delta$, so for $\alpha, \beta \geq 0$, $(d_u d_v)^\alpha \geq \delta^{2\alpha}$ and $(d_u + d_v)^\beta \geq (2\delta)^\beta$. For $\gamma \geq 0$ the exponential factor is at least 1 because the imbalance is non-negative, and multiplying and summing over the m edges yields (5.4). If $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ equals the bound for every graph, while otherwise equality requires equality in every active per-edge factor, which for $\alpha > 0$ or $\beta > 0$ forces $d_u = d_v = \delta$ on every edge. If $\alpha = \beta = 0$ and $\gamma > 0$ it forces $d_u = d_v$ on every edge, so the connected graph G is regular, hence δ -regular. Conversely, on a δ -regular graph every edge has degree pair (δ, δ) and equality holds. ■

The next proposition gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below. The bound is in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Proposition 5.8 (Ratio bound). *Let G be a simple graph with maximum degree at most Δ . Let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a BID index with edge contribution $h(i, j)$ strictly positive on every degree pair (i, j) that*

appears in G . Define

$$c_{\min}^h = \min_{(i,j)} \frac{\psi(i,j;\alpha,\beta,\gamma)}{h(i,j)}, \quad c_{\max}^h = \max_{(i,j)} \frac{\psi(i,j;\alpha,\beta,\gamma)}{h(i,j)},$$

where the extrema range over the degree pairs actually appearing in G . Then

$$c_{\min}^h \cdot I_h(G) \leq LO(G;\alpha,\beta,\gamma) \leq c_{\max}^h \cdot I_h(G), \quad (5.5)$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h .

Proof. On every edge uv we have $h(d_u, d_v) > 0$ and $c_{\min}^h \leq \psi(d_u, d_v; \alpha, \beta, \gamma)/h(d_u, d_v) \leq c_{\max}^h$. Multiplying by $h(d_u, d_v)$ and summing over $E(G)$, with $\sum_{uv} h(d_u, d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$, gives (5.5). Each side is an equality if and only if every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Proposition 5.8 is a *sharp graph-specific* comparison, whose constants depend on the degree pairs that actually appear in G , and a uniform version follows immediately by enlarging the range of the extrema.

Corollary 5.9 (Uniform ratio bound). *Assume the hypotheses of Proposition 5.8, but with h strictly positive on all of $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ and the extrema taken over P_Δ . Then inequality (5.5) holds simultaneously for every graph of maximum degree at most Δ , with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ . The uniform constants are in general strictly worse than the graph-specific ones, since the extrema are taken over a superset.*

6 QSPR diagnostics

We benchmark the Loyola family against the classical indices it reduces to on the 18 octane isomers, following the chemometric benchmark of the diminished Sombor study [28], and use the 35 nonane isomers as a replication

set with models refitted on it. The comparison set consists of the exact reductions of Section 1, $M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$, and the two fixed pure- γ benchmark points $LO(0, 0, 1)$ and $LO(0, 0, 2)$. Since the edge count $m = LO(0, 0, 0)$ is constant on the octanes, it is omitted. No parameter in the correlation comparison is tuned against the property data, and an exploratory local-tuning check (Supplementary Section S2) did not exceed the best fixed benchmark except by a 0.001 in-sample ΔH_{vap} margin. All comparisons are descriptive. On sets this small they indicate how imbalance information changes the fit of the parent family, and they do not establish chemical predictive utility. The pre-registered tests on measured boiling points of Section 6.4 are the primary predictive evidence of this paper, and the ridge regression on the degree-pair counts reported there is the reference against which every index model is read. The octane correlations of Section 6.1 are exploratory on eighteen molecules.

6.1 Correlations on the octane isomers

Table 4 reports the Pearson correlation r between each index and the boiling point T_B , the enthalpy of formation ΔH_f , the enthalpy of vaporisation ΔH_{vap} , the entropy S and the acentric factor ω of the 18 octane isomers. The property values are the standard octane benchmark compilation, audited value by value against the NIST WebBook [25] and the 1947 API Research Project 44 tables [40] (Supplementary Table S5 and Section S1). The open items are six entropies, the compiled acentric factors and the vaporisation enthalpy of 2,2,3,3-tetramethylbutane. That compound is crystalline at 298 K. The benchmark value 8.41 has no stated source and is consistent with a hypothetical supercooled liquid, for which sublimation minus fusion enthalpy gives about 8.56. The NIST value of 10.26 at 298 K describes sublimation (Supplementary Section S8). The entropies can move the 0.001 margin between HM and M_1 in the S column. In the ΔH_{vap} column $LO(0, 0, 1)$ stays first with the isomer excluded ($|r| = 0.980$) and with the hypothetical-liquid estimate ($|r| = 0.976$). Substituting the sublimation enthalpy, which mixes two processes in one column, lowers every $|r|$ to 0.374 or below. The ΔH_{vap} correlations of Table 4 are therefore conditional on a liquid-phase value for this isomer. We report the ordering

of the S column without drawing a conclusion from it. None of the open items enters the pre-registered tests of Sections 6.3 and 6.4.

Table 4. Pearson correlation r between the Loyola reductions, the fixed pure- γ benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the largest $|r|$ per column, and this is descriptive. Margins to the runner-up are 0.001–0.032, and their paired bootstrap intervals include zero (see the text).

Index	T_B	ΔH_f	ΔH_{vap}	S	ω
$M_1 = LO(0, 1, 0)$	−0.718	−0.762	−0.935	−0.970	−0.970
$M_2 = LO(1, 0, 0)$	−0.497	−0.542	−0.811	−0.934	−0.988
$HM = LO(0, 2, 0)$	−0.654	−0.710	−0.903	−0.971	−0.982
${}^mM_2 = LO(−1, 0, 0)$	+0.855	+0.891	+0.925	+0.863	+0.794
$R = LO(−1/2, 0, 0)$	+0.819	+0.850	+0.958	+0.932	+0.898
$\chi = LO(0, −1/2, 0)$	+0.800	+0.832	+0.961	+0.947	+0.924
$H/2 = LO(0, −1, 0)$	+0.821	+0.849	+0.961	+0.932	+0.900
$ISI = LO(1, −1, 0)$	−0.005	−0.000	−0.383	−0.593	−0.738
$GA = 2 LO(1/2, −1, 0)$	+0.822	+0.858	+0.965	+0.939	+0.906
$AG = \frac{1}{2} LO(−1/2, 1, 0)$	−0.816	−0.859	−0.957	−0.938	−0.900
$LO(0, 0, 1)$	−0.828	−0.832	−0.984	−0.928	−0.929
$LO(0, 0, 2)$	−0.829	−0.846	−0.981	−0.937	−0.925

The best correlation per property is split. ${}^mM_2 = LO(−1, 0, 0)$ is strongest on T_B and ΔH_f , $HM = LO(0, 2, 0)$ on S and $M_2 = LO(1, 0, 0)$ on ω , while the fixed pure- γ point $LO(0, 0, 1)$ is strongest on ΔH_{vap} ($|r| = 0.984$, with $LO(0, 0, 2)$ second at 0.981). Among the fixed indices compared, the pure- γ point is therefore the best single descriptor of the vaporisation enthalpy on this benchmark, and Section 6.3 shows that it also matches the tuned models. At the exact-reduction triples the Loyola correlations coincide with the classical indices, as they must. Recomputing each bolded correlation with single isomers deleted shifts $|r|$ by at most 0.027 (Supplementary Section S1). The bolded entries are the largest of twelve highly correlated sample correlations on 18 molecules and are therefore optimistic estimates of the selected descriptor’s correlation on new molecules. They are descriptive rankings rather than tests. Fourteen of the descriptor pairs have $|r| \geq 0.99$ and the first two principal components carry 99.05% of the variance (Supplementary Section S1), and the pure- γ points are collinear with the classical reductions (maximum $|r|$

with a reduction equal to 0.982 for $LO(0, 0, 1)$ and 0.993 for $LO(0, 0, 2)$. Differences in the third decimal of $|r|$ between near-collinear descriptors are therefore not meaningful rankings.

Proposition 4.1 is realised on these data, where the 18 isomers realise only 16 distinct edge-degree-pair count vectors. The two collision pairs of Table 1 receive the same value from every vertex-degree-based index at every parameter choice, although their measured properties differ, by up to 1.2°C in T_B within the first pair and 2.1°C within the second. Every correlation in Table 4, and every tuned variant below, operates above this irreducible floor.

6.2 Effect of freeing γ and of the search box

We ask whether, at identical search budgets, freeing γ improves out-of-sample prediction beyond an optimised parent $M_{\alpha,\beta}$. The design is a fully nested leave-one-out (Supplementary Section S3). For each seed, a set of candidate parameter vectors is drawn uniformly per coordinate from a search box and shared across properties and outer folds, triples (α, β, γ) for the Loyola model and pairs (α, β) with $\gamma = 0$ for the parent, with identical budget. Candidates are scored on the training isomers only, by inner leave-one-out root-mean-square error of the one-descriptor linear regression, after which the best candidate is refit and the held-out isomer predicted. No quantity computed on the held-out isomer enters selection or fitting.

In the box $(-2, 2)$ used first, freeing γ appeared to help the vaporisation enthalpy. It improved the nested Q^2 for ΔH_{vap} in 100/100 candidate streams at budgets 200 and 500, with median $\Delta Q^2 = +0.069$ and $+0.068$ (Supplementary Table S3), while it usually degraded ΔH_f and gave mixed results for the other properties. With 2,2,3,3-tetramethylbutane excluded the count falls to 93/100 and 96/100. These counts describe optimiser variance on one fixed set of 18 molecules, and they hide a boundary effect. For ΔH_{vap} the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds.

The decisive design choice is the box, which the nonane pre-registration also fixed at half-width 2. By $\ln(ij) = 2\ln(i + j) + \ln(1 - q^2) - \ln 4$

(Eq. (3.2)) the parent reaches imbalance information through $(1 - q^2)^\alpha$ along the direction $\beta \approx -2\alpha$, which needs large exponents and is exactly the direction the box truncates. On a dense parent grid (steps 0.05, 0.1, 0.2 and 0.25) the nested Q^2 of the parent for octane ΔH_{vap} is 0.8873, 0.9156, 0.9525 and 0.9383 at half-widths 2, 4, 8 and 12. The selected point lies within one grid step of the boundary in 18, 18, 18 and 0 of the 18 folds. On the nonanes the values are 0.8532, 0.8779, 0.8997 and 0.8837, with 34, 34, 34 and 0 boundary folds of 34. The half-width-12 box, chosen after inspecting these two ΔH_{vap} curves as the smallest examined in which the parent’s selections are interior, is used below for every property. The matched comparison was therefore repeated with 2000 candidates per model, seeds 0–49, and boxes of half-width 2 and 12 for both models (Supplementary Table S6); this repetition was not pre-registered. In that table, Edge is the share of outer folds, pooled over the 50 seeds, in which the parent’s selection lies within 10% of the boundary of the box. The counts are the number of seeds in which LO beats GM (half-width 2) or LO_h beats GM (half-width 12). In the wide box the vaporisation-enthalpy gain shrinks to about zero. A hybrid control widens only (α, β) and keeps $\gamma \in (-2, 2)$; it is drawn after the parent from the same candidate stream, so the two are paired by seed rather than by candidate. It gives median $\Delta Q^2 = +0.004$ on the octanes (28 of 50 seeds) and -0.002 on the nonanes (24 of 50), while widening all three coordinates gives -0.010 and -0.016 . On the other properties no advantage reaches the seed-count criterion in either box. Nonane T_B has 30 and 31 wins of 50 in the two boxes, and octane S moves from 11 to 38, which illustrates optimiser variance rather than an effect. Most of the change comes from the parent improving in the wide box ($0.887 \rightarrow 0.942$ on the octanes), and the rest from the three-dimensional search being slightly less efficient there ($0.957 \rightarrow 0.946$ for the hybrid control). What remains ($+0.004$ in 28 of 50 seeds) is within optimiser variance. The fixed point $LO(0, 0, 1)$ and the parameter-free *IRLA* match the fixed parent point $M_{4.5, -9}$ (leave-one-out $Q^2 = 0.957$ and 0.948 against 0.949 , on the octanes), which lies near the parent’s wide-box selections.

Under these protocols tuning γ yields no demonstrated predictive ad-

vantage over a parent evaluated over the wider box. Two caveats remain. Equal candidate budgets do not guarantee equally thorough searches in two and three dimensions, and the hybrid control reduces this asymmetry without removing it, so the matched design costs the three-parameter model part of the effect sought. The parent’s own Q^2 also varies with the box by about as much as the narrow-box gain, as the grid values above show. We conclude that no gain has been demonstrated, which leaves open whether a gain exists.

6.3 Replication on the nonane isomers

Under a protocol committed to the reproducibility package (commit 964b127) before any nonane value was collected, the identical nested design was applied to the 35 constitutional nonane isomers. Boiling points were 35 values from the NIST WebBook (NIST averages where given, otherwise the most recent determination, a deviation recorded in the protocol). Standard vaporisation enthalpies were 34 values from the NIST WebBook [25], of which 29 come from two compilations, the 1961 tabulation of Labbauf et al. [24] and the 1971 Wilhoit–Zwolinski handbook [45]. Agreement with them therefore measures consistency with compiled values rather than independent experimental validation, and, if the compiled values rest on structural increments, it favours any model linear in the degree-pair counts (Supplementary Section S4). The pre-registered verdict is that the γ effect *replicates* for ΔH_{vap} , with LO better than GM in 95/100 and 98/100 seeds and median $\Delta Q^2 = +0.048$ and $+0.053$, and is *mixed* for T_B (58/100 and 63/100). Within-fold baselines are in Supplementary Table S7. The protocol, however, fixed the box of half-width 2, which binds the parent (Supplementary Table S6), and in a box that does not, the nonane gain disappears (median $\Delta Q^2 = -0.002$ under the hybrid control). The pre-registered verdict is reported as obtained, with this deviation recorded. The seed-count criterion of that protocol counts optimiser variance on fixed data, and we now regard it as a design error of the protocol rather than as evidence. Because the nonane vaporisation enthalpies are compilation values, we decided after this result to treat the measured boiling points of Section 6.4 as the

primary predictive evidence of the paper and the nonane analysis as a secondary diagnostic.

On both sets the untuned fixed point $LO(0, 0, 1)$ matches or beats the tuned Loyola model (0.957 and 0.903 against medians 0.953 and 0.900). It is closely related to *IRLA*, since $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2} IRLA(G) + O(\gamma^2)$ as $\gamma \rightarrow 0$. At $\gamma = 1$ higher moments of the imbalance remain and trees with equal *IRLA* can differ in $LO(0, 0, 1)$, yet the correlation between the two is 0.996 on both sets, and *IRLA* itself comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900). What matters for this property is therefore a term linear in the imbalance. Ridge regression on the full degree-pair count vector, a reference for what the counts carry rather than a competing index, predicts the nonanes better than every index model ($Q^2 = 0.927$ for ΔH_{vap} , 0.682 for T_B). A post hoc molecule bootstrap conditional on the fitted predictions, which omits refitting and selection uncertainty, gives a 95% interval for LO minus GM on nonane ΔH_{vap} of $[-0.006, +0.113]$ at budget 200, which includes zero (Supplementary Section S4).

6.4 Measured boiling points

Measured vaporisation enthalpies are too scarce for a larger test. NIST lists only 25 acyclic alkane skeletons outside C_8 and C_9 , and one of the 75 decanes, with at least one calorimetric standard-state 298 K determination. Two further tests therefore used measured boiling points from the NIST WebBook [25], extracted by a fixed rule committed to the repository (commit `a5cf378`) before any model was fitted. The rule admits all hexanes, heptanes and octanes, 34 of the 35 nonanes and 34 of the 75 decanes. The included and excluded decanes do not differ detectably in branching or imbalance (Supplementary Section S9), but with 34 and 41 molecules only large differences are detectable, so the decane cohort is not shown to be representative. Test A evaluates the models on the decanes as a new cohort, refitting and reselecting inside each leave-one-out fold. Test B pools all 100 molecules and adds the carbon number n_C as a covariate, so that each descriptor is judged on branching beyond size. Because the octane and nonane boiling points had already been used in Sections 6.1 and 6.3,

Test B is a pooled size-adjusted analysis and not a fresh external test; only the hexanes, heptanes and decanes are new. Both use the non-truncating design of Section 6.2, with the parent searched over $(\alpha, \beta) \in (-12, 12)^2$ and the Loyola model over the same box with $\gamma \in (-2, 2)$, at 2000 candidates and seeds 0–49. The pre-registered criterion for a γ effect required at least 40 wins in 50 seeds and a molecule-bootstrap interval excluding zero (Table 5). The rule is one-sided, so a negative difference also returns “no evidence” and the rule never returns evidence that γ hurts.

Table 5. Measured boiling points (nested leave-one-out Q^2). Test A: decane cohort, refitted within folds. Test B: pooled C6–C10 with the carbon number n_C in every model (Class., *IRLA*, LO_1 : n_C plus that descriptor). Base: training-fold mean (Test A) or n_C alone (Test B). Class.: best classical index chosen in each fold. LO_1 : fixed $LO(0, 0, 1)$. Ridge: ridge regression on the degree-pair counts. GM, LO_h : medians over seeds 0–49 in the non-truncating boxes, LO_h being the hybrid control with (α, β) in the wide box and $\gamma \in (-2, 2)$. ΔQ^2 : median of LO_h minus GM. Last column: seeds (of 50) in which LO_h beats GM.

Test	Base	Class.	<i>IRLA</i>	LO_1	Ridge	GM	LO_h	ΔQ^2	>
decanes ($n = 34$)	−0.062	0.176	0.167	0.149	0.860	0.514	0.406	−0.072	15
C6–C10, with n_C ($n = 100$)	0.938	0.967	0.963	0.963	0.991	0.977	0.977	−0.000	25

Both pre-registered verdicts are negative. On the decane cohort the median $\Delta Q^2 = -0.072$, LO is ahead in 15 of 50 seeds, and the 95% interval is $[-0.156, +0.001]$. In the pooled size-adjusted set the values are -0.000 , 25 of 50 and $[-0.003, +0.002]$. In the original box of half-width 2, reported as the pre-registration requires, the two tuned models are level on the decanes (medians 0.469 and 0.465). The Loyola model is marginally ahead on the pooled set (0.980 against 0.976). The decane verdict therefore does not depend on the box. The lower decane Q^2 of the hybrid control (0.406 against 0.465 in the narrow box) is consistent with the cost of a three-dimensional search over a wide box; the one-sided design cannot separate that cost from a penalty of selecting γ . The pooled cohort was corrected after an external audit found one nonane’s second NIST entry skipped (a deviation recorded in the repository), pooling grouped pages also moved one heptane value by 0.2°C , and the Test B solver was re-conditioned after the first results (also recorded). On the uncorrected cohort of 101 molecules

the same test gave -0.000 , 24 of 50, $[-0.003, +0.002]$, so the correction changes no conclusion. The intervals are molecule bootstraps conditional on the fitted predictions, as in Section 6.3, and omit refitting and selection. An exploratory resampling added after the results reruns the whole nested procedure for both models on random subsamples of 80% of the molecules, each with a fresh candidate stream. On the decanes (500 subsamples) the median ΔQ^2 is -0.110 with central 95% range $[-0.510, +0.312]$. On the pooled set (200 subsamples) it is $+0.001$ with range $[-0.015, +0.008]$. Neither range supports a gain from γ , and the decane range shows that the conditional interval understates the uncertainty. Ridge regression on the ten degree-pair counts reaches $Q^2 = 0.860$ on the decanes, where every single index stays between 0.15 and 0.51, and 0.991 on the pooled set. A permutation check on the decanes repeats the whole nested procedure, including the choice of the ridge penalty in every fold, for 200 random permutations of the boiling points. The permuted Q^2 has median -0.09 , 95th percentile -0.01 and maximum 0.20. None reaches 0.860, so the empirical p -value is $1/201 \approx 0.005$, the smallest that 200 permutations can give. A permutation test cannot exclude every form of leakage. Branching effects on the boiling point are thus spread over several degree-pair counts in a way that no linear model on a single member of the family, with or without γ , recovers; by Proposition 4.2 the values themselves still separate the profiles. These conclusions concern acyclic alkanes.

7 Discussion and limitations

The results of Sections 3–5 are exact. Their limits, and those of the statistical results of Section 6, are as follows.

- (i) The minimality of the collision orders in Theorem 3.5(iii), equality in Proposition 3.13(ii) for $\Delta \leq 6$, and the recovery of discrimination off Π in Section 4.2 rest on exhaustive enumeration. No general argument is given.
- (ii) Whether two graphs with $\Delta \leq 4$ can exchange order five times along γ is open (Remark 3.12).

- (iii) The data sets are small: 18 octanes, 35 nonanes, 34 decanes and 100 pooled C6–C10 alkanes, all acyclic.
- (iv) Six octane entropies and the compiled acentric factors rest on secondary sources, and the vaporisation enthalpy of 2,2,3,3-tetramethylbutane has an unresolved reference state (Supplementary Section S8). The orderings of the S and ω columns of Table 4 are not used, and the magnitudes of its ΔH_{vap} correlations depend on that one value.
- (v) Equal candidate budgets do not give equally thorough searches in two and three dimensions. The hybrid control of Section 6.2 reduces this asymmetry without removing it.
- (vi) Changes made after outcomes had been seen are listed in Table 6. None changes a pre-registered verdict as reported, but the wide-box analysis that reverses the nonane conclusion is itself not pre-registered.

Table 6. Changes to the pre-registered protocols made after outcomes had been seen. Protocols and change records are in the reproducibility package [30].

Test		Change after outcomes	Effect
Nonanes tion 6.3)	(Sec-	Parent box widened from half-width 2 to 12, chosen after inspecting the ΔH_{vap} curves	Registered verdict reported as obtained; the gain disappears in the wide box
Nonanes		Seed-count criterion judged to count optimiser variance	None on reported numbers
Nonanes		Measured boiling points made the primary predictive evidence	None on reported numbers
Boiling Test B	points,	All NIST pages of a skeleton pooled; one nonane excluded, one heptane value moved by 0.2 °C	Verdict unchanged; both cohorts reported
Boiling Test B	points,	Least-squares solver rescaled by columns	Verdict unchanged
Boiling both	points,	Selection-aware resampling added (Section 6.4)	Exploratory; no verdict depends on it

8 Conclusion

We extended the Gutman–Milovanović family by an exponential factor in the normalised degree imbalance of each edge and asked what this third

coordinate adds. The answer on the mathematical side is exact. On the ratio plane the pure-imbalance points at nonzero algebraic γ discriminate the histogram of edge imbalances and nothing finer, a limit that binds every index depending on an edge only through its degree ratio. For trees this histogram carries the full degree-pair information up to order twelve and then loses it, at order thirteen in general and sixteen for chemical trees, with explicit colliding families. Along the imbalance axis two chemical graphs exchange order at most five times, the extremal trees of the pure-imbalance points are the path and the star, and at rational exponents a nonzero algebraic imbalance parameter never destroys a distinction the parent makes.

On the chemometric side no predictive gain was demonstrated. An apparent gain from the imbalance parameter on vaporisation enthalpies replicated under a pre-registered protocol, but it disappeared once the parent family was searched over a box that did not truncate it. Pre-registered tests on measured boiling points of decanes and of a pooled set of C6–C10 alkanes showed no demonstrated benefit from the third coordinate. There a linear model on the degree-pair counts outperformed every single index. The untuned pure-imbalance point is highly correlated with the parameter-free irregularity index and matched the tuned models on vaporisation enthalpy. The compactness of the family therefore comes from linear imbalance weighting rather than from tuning. The third coordinate remains useful as an interpretable direction on the parent family along which the effect of degree imbalance can be analysed exactly. Three problems remain open. The first is sharp two-sided bounds for the family in terms of indices outside it, such as the atom-bond-connectivity and Sombor indices. The second is the extremal graphs for general parameter triples, beginning with the maximal chemical trees of the pure-imbalance points, and for unicyclic and bicyclic graphs. The third is the chemometric behaviour on other compound classes, such as benzene derivatives, and on class-specific endpoints.

Data availability. Data, scripts and verifiers that recompute every tabulated value are in the reproducibility package [30], where the nonane and boiling-point protocols were committed before the corresponding models

were fitted. The termwise bounds, the convexity statement, the independence of the imbalance direction and the order-thirteen profile collision are machine-checked in Lean 4 with Mathlib (Supplementary Section S7), and the file `LEAN_VERIFICATION.md` of the repository maps each statement of this paper to its Lean theorem.

Acknowledgment: The authors thank colleagues for helpful discussions.

References

- [1] M. O. Albertson, The irregularity of a graph, *Ars Comb.* **46** (1997) 219–225.
- [2] A. Ali, I. Gutman, B. Furtula, Extremal results on bond incident degree indices of graphs: a survey, *Aequationes Math.* **100** (2026) #66.
- [3] A. Ali, I. Gutman, B. Furtula, A. M. Albalahi, A. E. Hamza, On chemical and mathematical characteristics of generalized degree-based molecular descriptors, *AIMS Math.* **10** (2025) 6788–6804.
- [4] A. Baker, *Transcendental Number Theory*, Cambridge Univ. Press, Cambridge, 1975.
- [5] J. Barman, S. Das, Geometric approach to degree-based topological index: Hyperbolic Sombor index, *MATCH Commun. Math. Comput. Chem.* **95** (2026) 63–94.
- [6] B. Bollobás, P. Erdős, Graphs of extremal weights, *Ars Comb.* **50** (1998) 225–233.
- [7] L. D. Brown, *Fundamentals of Statistical Exponential Families with Applications in Statistical Decision Theory*, IMS Lecture Notes–Monograph Series 9, Institute of Mathematical Statistics, Hayward, 1986.
- [8] R. Cruz, J. Monsalve, J. Rada, Trees with maximum exponential Randić index, *Discr. Appl. Math.* **283** (2020) 634–643.
- [9] R. Cruz, J. Rada, The path and the star as extremal values of vertex-degree-based topological indices among trees, *MATCH Commun. Math. Comput. Chem.* **82** (2019) 715–732.

-
- [10] K. C. Das, S. Mondal, On exponential geometric-arithmetic index of graphs, *J. Math. Chem.* **62** (2024) 2740–2760.
- [11] P. Das, S. Mondal, Z. Raza, A. Pal, On exponential inverse symmetric division deg index of graphs, *J. Appl. Math. Comput.* **71** (Suppl. 1) (2025) 537–561.
- [12] B. Furtula, I. Gutman, M. Dehmer, On structure-sensitivity of degree-based topological indices, *Appl. Math. Comput.* **219** (2013) 8973–8978.
- [13] W. Gao, Y. Gao, The extremal trees for exponential vertex-degree-based topological indices, *Appl. Math. Comput.* **472** (2024) #128634.
- [14] A. Granados, A. Portilla, Y. Quintana, E. Tourís, Bounds for the Gutman–Milovanović index and some applications, *J. Math. Chem.* **63** (2025) 406–418.
- [15] I. Gutman, Geometric approach to degree-based topological indices: Sombor indices, *MATCH Commun. Math. Comput. Chem.* **86** (2021) 11–16.
- [16] I. Gutman, Survey of Sombor-type degree-based topological indices, *MATCH Commun. Math. Comput. Chem.* **96** (2026) 819–837.
- [17] I. Gutman, J. Buragohain, G. G. Nyauli, Concealed relations between vertex-degree-based topological indices, *J. Math. Chem.* **64** (2026) #39.
- [18] I. Gutman, E. Milovanović, I. Milovanović, Beyond the Zagreb indices, *AKCE Int. J. Graphs Comb.* **17** (2020) 74–85.
- [19] I. Gutman, I. Redžepović, G. Özkan Kizilirmak, V. R. Kulli, Euler–Sombor index and its congeners, *Open J. Math. Sci.* **9** (2025) 141–148.
- [20] I. Gutman, N. Trinajstić, Graph theory and molecular orbitals. Total π -electron energy of alternant hydrocarbons, *Chem. Phys. Lett.* **17** (1972) 535–538.
- [21] Z. Hu, L. Li, X. Li, D. Peng, Extremal graphs for topological index defined by a degree-based edge-weight function, *MATCH Commun. Math. Comput. Chem.* **88** (2022) 505–520.
- [22] G. J. O. Jameson, Counting zeros of generalised polynomials: Descartes’ rule of signs and Laguerre’s extensions, *Math. Gaz.* **90** (2006) 223–234.

-
- [23] E. V. Konstantinova, The discrimination ability of some topological and information distance indices for graphs of unbranched hexagonal systems, *J. Chem. Inf. Comput. Sci.* **36** (1996) 54–57.
- [24] A. Labbauf, J. B. Greenshields, F. D. Rossini, Heats of formation, combustion, and vaporization of the 35 nonanes and 75 decanes, *J. Chem. Eng. Data* **6** (1961) 261–263.
- [25] P. J. Linstrom, W. G. Mallard (Eds.), *NIST Chemistry WebBook, NIST Standard Reference Database Number 69*, National Institute of Standards and Technology, Gaithersburg, 2025, <https://doi.org/10.18434/T4D303>.
- [26] B. S. Mityagin, The zero set of a real analytic function, *Math. Notes* **107** (2020) 529–530.
- [27] E. D. Molina, J. M. Rodríguez-García, J. M. Sigarreta, S. J. Torralbas Fitz, On the Gutman–Milovanović index and chemical applications, *AIMS Math.* **10** (2025) 1998–2020.
- [28] F. Movahedi, I. Gutman, I. Redžepović, B. Furtula, Diminished Sombor index, *MATCH Commun. Math. Comput. Chem.* **95** (2026) 141–162.
- [29] Z. Lin, T. Zhou, X. Wang, L. Miao, The general Albertson irregularity index of graphs, *AIMS Math.* **7** (2022) 25–38.
- [30] A. Natarajan, G. Shiwakoti, M. Arockiaraj, The Loyola index family: data, scripts and verifiers, <https://github.com/Singati2/LOYOLA-PAPER> (2026).
- [31] G. G. Nyauli, J. Buragohain, Geometric approach to degree-based topological indices: cosine-rule generalized Sombor indices, *MATCH Commun. Math. Comput. Chem.* **96** (2026) 295–324.
- [32] G. G. Nyauli, J. Buragohain, Geometric approach to degree-based topological indices: degree-ratio Sombor indices of graphs, *MATCH Commun. Math. Comput. Chem.* **97** (2027) 135–164.
- [33] A. Portilla, J. M. Rodríguez, I. Salas Lorenzo, J. M. Sigarreta, The Gutman–Milovanović index in mathematical chemistry, *J. Math. Chem.* **63** (2025) 2068–2087.
- [34] J. Rada, Exponential vertex-degree-based topological indices and discrimination, *MATCH Commun. Math. Comput. Chem.* **82** (2019) 29–41.

-
- [35] J. Rada, A finite-dimensional linear framework for vertex-degree-based topological indices, *MATCH Commun. Math. Comput. Chem.* **96** (2026) 841–863.
- [36] J. Rada, J. M. Rodríguez, J. M. Sigarreta, Weakly discriminating vertex-degree-based topological indices, *MATCH Commun. Math. Comput. Chem.* **87** (2022) 97–108.
- [37] M. Rakić, B. Furtula, A novel method for measuring the structure sensitivity of molecular descriptors, *J. Chemom.* **33** (2019) #e3138.
- [38] M. Randić, Characterization of molecular branching, *J. Am. Chem. Soc.* **97** (1975) 6609–6615.
- [39] T. Réti, R. Sharafadini, Á. Drégelyi-Kiss, H. Haghbin, Graph irregularity indices used as molecular descriptors in QSPR studies, *MATCH Commun. Math. Comput. Chem.* **79** (2018) 509–524.
- [40] F. D. Rossini, K. S. Pitzer, W. J. Taylor, J. P. Ebert, J. E. Kilpatrick, C. W. Beckett, M. G. Williams, H. G. Werner, *Selected Values of Properties of Hydrocarbons*, Natl. Bur. Stand. Circ. 461, U.S. Government Printing Office, Washington, 1947.
- [41] G. H. Shirdel, H. Rezapour, A. M. Sayadi, The hyper-Zagreb index of graph operations, *Iran. J. Math. Chem.* **4** (2013) 213–220.
- [42] J. M. Sigarreta, Extremal problems on exponential vertex-degree-based topological indices, *Math. Biosci. Eng.* **19** (2022) 6985–6995.
- [43] D. Vukičević, B. Furtula, Topological index based on the ratios of geometrical and arithmetical means of end-vertex degrees of edges, *J. Math. Chem.* **46** (2009) 1369–1376.
- [44] D. Vukičević, M. Gašperov, Bond additive modeling 1. Adriatic indices, *Croat. Chem. Acta* **83** (2010) 243–260.
- [45] R. C. Wilhoit, B. J. Zwolinski, *Handbook of Vapor Pressures and Heats of Vaporization of Hydrocarbons and Related Compounds*, Thermodynamics Research Center, Texas A&M University, College Station, 1971.
- [46] B. Zhou, N. Trinajstić, On a novel connectivity index, *J. Math. Chem.* **46** (2009) 1252–1270.
- [47] B. Zhou, N. Trinajstić, On general sum-connectivity index, *J. Math. Chem.* **47** (2010) 210–218.